



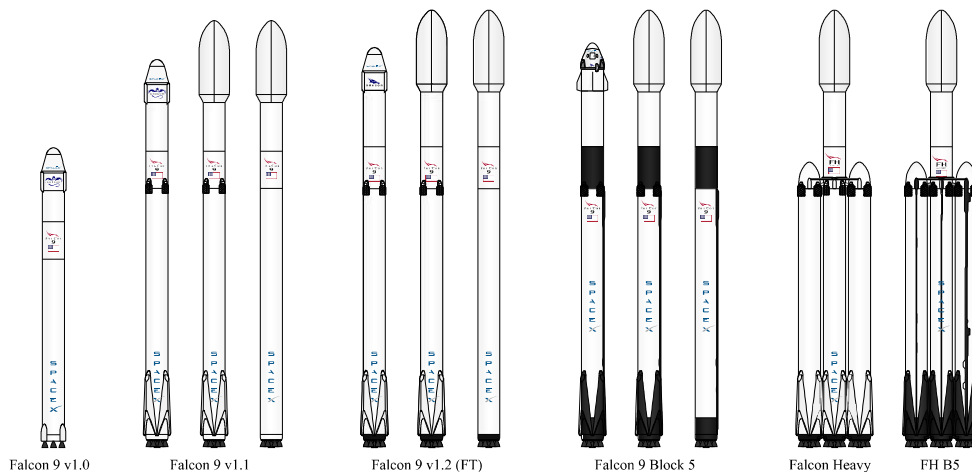
Space X Falcon 9 First Stage Landing Prediction

Web scraping Falcon 9 and Falcon Heavy Launches Records from Wikipedia

Estimated time needed: **40** minutes

In this lab, you will be performing web scraping to collect Falcon 9 historical launch records from a Wikipedia page titled `List of Falcon 9 and Falcon Heavy launches`

https://en.wikipedia.org/wiki/List_of_Falcon_9_and_Falcon_Heavy_launches



Falcon 9 first stage will land successfully



Several examples of an unsuccessful landing are shown here:



More specifically, the launch records are stored in a HTML table shown below:

2020 [edit]

In late 2019, Gwynne Shotwell stated that SpaceX hoped for as many as 24 launches for Starlink satellites in 2020,^[400] in addition to 14 or 15 non-Starlink launches. At 26 launches, 13 of which for Starlink satellites, Falcon 9 had its most prolific year, and Falcon rockets were second most prolific rocket family of 2020, only behind China's Long March rocket family.^[401]

[190] Flight No.	Date and time (UTC)	Version, Booster ^[2]	Launch site	Payload ^[4]	Payload mass	Orbit	Customer	Launch outcome	Booster landing
78	7 January 2020, 02:19:21 ^[392]	F9 B5 Δ, B1049.4	CCAFS, SLC-40	Starlink 2 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[3]	LEO	SpaceX	Success	Success (shore ship)
Third large batch and second operational flight of Starlink constellation. One of the 60 satellites included a test coating to make the satellite less reflective, and thus less likely to interfere with ground-based astronomical observations. ^[406]									
79	19 January 2020, 15:30 ^[404]	F9 B5 Δ, B1046.4	KSC, LC-39A	Crew Dragon in-flight abort test ^[405] (Dragon C205.1)	12,050 kg (26,570 lb)	Sub-orbital ^[406]	NASA (CTS) ^[407]	Success	No attempt
An atmospheric test of the Dragon 2 abort system after Max Q. The capsule fired its SuperDraco engines, reached an apogee of 49 km (25 mi), deployed parachutes after reentry, and splashed down in the ocean 31 km (19 mi) downrange from the launch site. The test was previously slated to be accomplished with the Crew Dragon Demo-1 capsule, ^[408] but that test article exploded during a ground test of SuperDraco engines on 20 April 2019. ^[423] The abort test used the capsule originally intended for the first crewed flight. ^[394] As expected, the booster was destroyed by aerodynamic forces after the capsule aborted. ^[404] First flight of a Falcon 9 with only one functional stage — the second stage had a mass simulator in place of its engine.									
80	29 January 2020, 14:27 ^[401]	F9 B5 Δ, B1051.3	CCAFS, SLC-40	Starlink 3 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[3]	LEO	SpaceX	Success	Success (shore ship)
Third operational and fourth large batch of Starlink satellites, deployed in a circular 290 km (180 mi) orbit. One of the faring halves was caught, while the other was fished out of the ocean. ^[402]									
81	17 February 2020, 15:00 ^[403]	F9 B5 Δ, B1056.4	CCAFS, SLC-40	Starlink 4 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[3]	LEO	SpaceX	Success	Failure (shore ship)
Fourth operational and fifth large batch of Starlink satellites. Used a new flight profile which deployed into a 212 km × 386 km (132 mi × 240 mi) elliptical orbit instead of launching into a circular orbit and firing the second stage engine twice. The first stage booster failed to land on the drone ship ^[404] due to incorrect wind data. ^[405] This was the first time a flight proven booster failed to land.									
82	7 March 2020, 04:50 ^[406]	F9 B5 Δ, B1059.2	CCAFS, SLC-40	Spacex CRS-20 (Dragon C112.3 Δ)	1,977 kg (4,359 lb) ^[407]	LEO (ISS)	NASA (CRS)	Success	Success (ground pad)
Last launch of phase 1 of the CRS contract. Carries Bartolomeo, an ESA platform for hosting external payloads onto ISS. ^[408] Originally scheduled to launch on 2 March 2020, the launch date was pushed back due to a second stage engine failure. SpaceX decided to swap out the second stage instead of replacing the faulty part. ^[409] It was SpaceX's 50th successful landing of a first stage booster, the third flight of the Dragon C112 and the last launch of the cargo Dragon spacecraft.									
83	18 March 2020, 12:16 ^[410]	F9 B5 Δ, B1048.5	KSC, LC-39A	Starlink 5 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[3]	LEO	SpaceX	Success	Failure (shore ship)
Fifth operational launch of Starlink satellites. It was the first time a first stage booster flew for a fifth time and the second time the fairings were reused (Starlink flight in May 2019). ^[411] Towards the end of the first stage burn, the booster suffered premature shut down of an engine, the first of a Merlin 1D variant and first since the CRS-1 mission in October 2012. However, the payload still reached the targeted orbit. ^[412] This was the second Starlink launch booster landing failure in a row, later revealed to be caused by residual cleaning fluid trapped inside a sensor. ^[413]									
84	22 April 2020, 19:30 ^[414]	F9 B5 Δ, B1051.4	KSC, LC-39A	Starlink 6 v1.0 (60 satellites)	15,600 kg (34,400 lb) ^[3]	LEO	SpaceX	Success	Success (shore ship)

Objectives

Web scrap Falcon 9 launch records with `BeautifulSoup` :

- Extract a Falcon 9 launch records HTML table from Wikipedia
- Parse the table and convert it into a Pandas data frame

First let's import required packages for this lab

```
In [1]: !pip3 install beautifulsoup4
!pip3 install requests
```

```
Requirement already satisfied: beautifulsoup4 in /home/jupyterlab/conda/envs/python/
lib/python3.7/site-packages (4.11.1)
Requirement already satisfied: soupsieve>1.2 in /home/jupyterlab/conda/envs/python/l
ib/python3.7/site-packages (from beautifulsoup4) (2.3.2.post1)
Requirement already satisfied: requests in /home/jupyterlab/conda/envs/python/lib/py
thon3.7/site-packages (2.29.0)
Requirement already satisfied: charset-normalizer<4,>=2 in /home/jupyterlab/conda/en
vs/python/lib/python3.7/site-packages (from requests) (3.1.0)
Requirement already satisfied: idna<4,>=2.5 in /home/jupyterlab/conda/envs/python/li
b/python3.7/site-packages (from requests) (3.4)
Requirement already satisfied: urllib3<1.27,>=1.21.1 in /home/jupyterlab/conda/envs/
python/lib/python3.7/site-packages (from requests) (1.26.15)
Requirement already satisfied: certifi>=2017.4.17 in /home/jupyterlab/conda/envs/pyt
hon/lib/python3.7/site-packages (from requests) (2023.5.7)
```

```
In [2]: import sys

import requests
from bs4 import BeautifulSoup
import re
import unicodedata
import pandas as pd
```

and we will provide some helper functions for you to process web scraped HTML table

```
In [3]: def date_time(table_cells):
    """
    This function returns the data and time from the HTML table cell
    Input: the element of a table data cell extracts extra row
    """
    return [data_time.strip() for data_time in list(table_cells.strings)][0:2]

def booster_version(table_cells):
    """
    This function returns the booster version from the HTML table cell
    Input: the element of a table data cell extracts extra row
    """
    out=''.join([booster_version for i,booster_version in enumerate(table_cells.st
    return out

def landing_status(table_cells):
```

```

"""
This function returns the landing status from the HTML table cell
Input: the element of a table data cell extracts extra row
"""
out=[i for i in table_cells.strings][0]
return out

def get_mass(table_cells):
    mass=unicodedata.normalize("NFKD", table_cells.text).strip()
    if mass:
        mass.find("kg")
        new_mass=mass[0:mass.find("kg")+2]
    else:
        new_mass=0
    return new_mass

def extract_column_from_header(row):
    """
    This function returns the landing status from the HTML table cell
    Input: the element of a table data cell extracts extra row
    """
    if (row.br):
        row.br.extract()
    if row.a:
        row.a.extract()
    if row.sup:
        row.sup.extract()

    column_name = ' '.join(row.contents)

    # Filter the digit and empty names
    if not(column_name.strip().isdigit()):
        column_name = column_name.strip()
        return column_name

```

To keep the lab tasks consistent, you will be asked to scrape the data from a snapshot of the [List of Falcon 9 and Falcon Heavy launches](#) Wikipage updated on [9th June 2021](#)

```

In [4]: static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falco
headers = {
    "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
                "AppleWebKit/537.36 (KHTML, like Gecko) "
                "Chrome/91.0.4472.124 Safari/537.36"
}

```

Next, request the HTML page from the above URL and get a `response` object

TASK 1: Request the Falcon9 Launch Wiki page from its URL

First, let's perform an HTTP GET method to request the Falcon9 Launch HTML page, as an HTTP response.

```
In [5]: # use requests.get() method with the provided static_url and headers
# assign the response to a object
# Request the page
response = requests.get(static_url, headers=headers)
```

Create a `BeautifulSoup` object from the HTML `response`

```
In [6]: # Use BeautifulSoup() to create a BeautifulSoup object from a response text content
soup = BeautifulSoup(response.text, "html.parser")
```

Print the page title to verify if the `BeautifulSoup` object was created properly

```
In [7]: # Use soup.title attribute
print(soup.title)
```

```
<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>
```

TASK 2: Extract all column/variable names from the HTML table header

Next, we want to collect all relevant column names from the HTML table header

Let's try to find all tables on the wiki page first. If you need to refresh your memory about `BeautifulSoup`, please check the external reference link towards the end of this lab

```
In [8]: # Use the find_all function in the BeautifulSoup object, with element type `table`
# Assign the result to a list called `html_tables`
html_tables = soup.find_all('table')
```

Starting from the third table is our target table contains the actual launch records.

```
In [9]: # Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
```

```

<table class="wikitable plainrowheaders collapsible" id="mwmQ" style="width: 100%;">
<tbody id="mwmG"><tr id="mwmw"><th id="mwnA" scope="col">Flight No.</th>
<th id="mwnQ" scope="col">Date and<br id="mwng"/>time (<a href="https://en.wikipedi
a.org/wiki/Coordinated_Universal_Time" id="mwnw" rel="mw:WikiLink" title="Coordinate
d Universal Time">UTC</a></th>
<th id="mwoA" scope="col"><a href="https://en.wikipedia.org/wiki/List_of_Falcon_9_fi
rst-stage_boosters" id="mwoQ" rel="mw:WikiLink" title="List of Falcon 9 first-stage
boosters">Version,<br id="mwog"/>Booster</a> <sup about="#mwt70" class="mw-ref refer
ence" data-mw="{"name":"ref","attrs":{"name":"booster","group":"lower-alpha"},"bod
y":{"id":"mw-reference-text-cite_note-booster-11","parts":[{"template":{"target":
{"wt":"efn","href":"./Template:Efn"},"params":{"name":{"wt":"booster"},"1":{"wt":"Fa
lcon 9 first-stage boosters are designated with a construction serial number and an
optional flight number when reused, e.g. B1021.1 and B1021.2 represent the two fligh
ts of booster [[B1021]]. Launches using reused boosters are denoted with a recycled
symbol ♻️.}}},"i":0}}}}" id="cite_ref-booster_11-0" rel="dc:references" typeof="mw:Tr
ansclusion mw:Extension/ref"><a data-mw-group="lower-alpha" href="#cite_note-booster
-11" id="mwow"><span class="mw-reflink-text" id="mwpA"><span class="cite-bracket" id
="mwpQ"></span><b><span class="cite-bracket" id="mwpG"></span></b></span></a></sup></th>
<th id="mwpw" scope="col">Launch site</th>
<th id="mwqA" scope="col">Payload<sup about="#mwt73" class="mw-ref reference" data-m
w="{"name":"ref","attrs":{"name":"Dragon","group":"lower-alpha"},"body":{"id":"mw-re
ference-text-cite_note-Dragon-12","parts":[{"template":{"target":{"wt":"efn","hre
f":"./Template:Efn"},"params":{"name":{"wt":"Dragon"},"1":{"wt":"Dragon 1 or 2 are d
esignated with a construction serial number or name and an optional flight number wh
en reused, e.g. Dragon C106.1 and Dragon C106.2 represent the two flights of [[Drago
n C106]]. Dragon Spacecrafts that are reused are denoted with a recycled symbol
♻️.}}},"i":0}}}}" id="cite_ref-Dragon_12-0" rel="dc:references" typeof="mw:Transclusi
on mw:Extension/ref"><a data-mw-group="lower-alpha" href="#cite_note-Dragon-12" id
="mwqQ"><span class="mw-reflink-text" id="mwqG"><span class="cite-bracket" id="mwq
W"></span><b><span class="cite-bracket" id="mwrA"></span></b></span></a></sup></th>
<th id="mwrQ" scope="col">Payload mass</th>
<th id="mwrG" scope="col">Orbit</th>
<th id="mwrw" scope="col">Customer</th>
<th id="mwsA" scope="col">Launch<br id="mwsQ"/>outcome</th>
<th id="mwsg" scope="col"><a href="https://en.wikipedia.org/wiki/Falcon_9_first-stag
e_landing_tests" id="mwsw" rel="mw:WikiLink" title="Falcon 9 first-stage landing tes
ts">Booster<br id="mwTA"/>landing</a></th></tr>
<tr id="mwtQ">
<th id="mwtG" rowspan="2" scope="row" style="text-align:center;">1</th>
<td id="mwtw">4 June 2010,<br id="mwuA"/>18:45</td>
<td id="mwuQ"><a href="https://en.wikipedia.org/wiki/Falcon_9_v1.0" id="mwug" rel="m
w:WikiLink" title="Falcon 9 v1.0">F9 v1.0</a><sup about="#mwt78" class="mw-ref refer
ence" data-mw="{"name":"ref","attrs":{"name":"MuskMay2012"},"body":{"id":"mw-referen
ce-text-cite_note-MuskMay2012-13"}}" id="cite_ref-MuskMay2012_13-0" rel="dc:referenc
es" typeof="mw:Extension/ref"><a href="#cite_note-MuskMay2012-13" id="mwuw"><span cl
ass="mw-reflink-text" id="mwvA"><span class="cite-bracket" id="mwvQ"></span>7<span
class="cite-bracket" id="mwvG"></span></span></a></sup><br id="mwvw"/>B0003.1<sup a
bout="#mwt79" class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"block
_numbers"}}" id="cite_ref-block_numbers_14-0" rel="dc:references" typeof="mw:Extensi
on/ref"><a href="#cite_note-block_numbers-14" id="mwWA"><span class="mw-reflink-tex
t" id="mwWQ"><span class="cite-bracket" id="mwWG"></span>8<span class="cite-bracke
t" id="mwWW"></span></span></a></sup></td>
<td id="mwxA"><a href="https://en.wikipedia.org/wiki/Cape_Canaveral_Space_Force_Stat
ion" id="mwXQ" rel="mw:WikiLink" title="Cape Canaveral Space Force Station">CCAFS</a
>,<br id="mwXG"/><a href="https://en.wikipedia.org/wiki/Cape_Canaveral_Space_Launch_
Complex_40" id="mwXW" rel="mw:WikiLink" title="Cape Canaveral Space Launch Complex 4

```

```

0">SLC-40</a></td>
<td id="mwYQ"><a href="https://en.wikipedia.org/wiki/Dragon_Spacecraft_Qualification_Unit" id="mwYQ" rel="mw:WikiLink" title="Dragon Spacecraft Qualification Unit">Dragon Spacecraft Qualification Unit</a></td>
<td id="mwYg"></td>
<td id="mwYw"><a href="https://en.wikipedia.org/wiki/Low_Earth_orbit" id="mwZA" rel="mw:WikiLink" title="Low Earth orbit">LEO</a></td>
<td id="mwZQ"><a href="https://en.wikipedia.org/wiki/SpaceX" id="mwzg" rel="mw:WikiLink" title="SpaceX">SpaceX</a></td>
<td about="#mwT80" class="table-success" data-mw="{"parts":["|","template":{"target":{"wt":"Success","href":"./Template:Success"},"params":{},"i":0}}}" id="mwzw" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center; " typeof="mw:Transclusion">Success</td>
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<sup about="#mwT84" class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"ns20110930"},"body":{"i":"mw-reference-text-cite_note-ns20110930-15"}}}" id="cite_ref-ns20110930_15-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-ns20110930-15" id="mw0Q"><span class="mw-reflink-text" id="mw0g"><span class="cite-bracket" id="mw0w">[</span>9<span class="cite-bracket" id="mw1A"></span></span></a></sup><sup about="#mwT87" class="mw-ref reference" data-mw="{"name":"ref","attrs":{},"body":{"i":"mw-reference-text-cite_note-16"}}}" id="cite_ref-16" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-16" id="mw1Q"><span class="mw-reflink-text" id="mw1g"><span class="cite-bracket" id="mw1w">[</span>10<span class="cite-bracket" id="mw2A"></span></span></a></sup><br/><small>(parachute)</small></td></tr>
<tr id="mw2Q">
<td colspan="9" id="mw2g">First flight of Falcon 9 v1.0.<sup about="#mwT88" class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"sfn20100604"}}}" id="cite_ref-sfn20100604_17-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-sfn20100604-17" id="mw2w"><span class="mw-reflink-text" id="mw3A"><span class="cite-bracket" id="mw3Q">[</span>11<span class="cite-bracket" id="mw3g"></span></span></a></sup> Used a boilerplate version of Dragon capsule which was not designed to separate from the second stage.<small id="mw3w">(<a class="mw-selflink-fragment" href="#First_flight_of_Falcon_9" id="mw4A" rel="mw:WikiLink">more details below</a>)</small> Attempted to recover the first stage by parachuting it into the ocean, but it burned up on reentry, before the parachutes even deployed.<sup about="#mwT91" class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"parachute"},"body":{"i":"mw-reference-text-cite_note-parachute-18"}}}" id="cite_ref-parachute_18-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-parachute-18" id="mw4Q"><span class="mw-reflink-text" id="mw4g"><span class="cite-bracket" id="mw4w">[</span>12<span class="cite-bracket" id="mw5A"></span></span></a></sup></td></tr>
<tr id="mw5Q">
<th id="mw5g" rowspan="2" scope="row" style="text-align:center;">2</th>
<td id="mw5w">8 December 2010,<br id="mw6A"/>15:43<sup about="#mwT94" class="mw-ref

```

```

reference" data-mw="{ "name": "ref", "attrs": { "name": "spaceflightnow Clark Launch Report-19" } }' id="cite_ref-spaceflightnow_Clark_Launch_Report-19-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-spaceflightnow_Clark_Launch_Report-19" id="mw6Q"><span class="mw-reflink-text" id="mw6g"><span class="cite-bracket" id="mw6w"></span>13<span class="cite-bracket" id="mw7A"></span></span></a></sup></td>
<td id="mw7Q"><a href="https://en.wikipedia.org/wiki/Falcon_9_v1.0" id="mw7g" rel="mw:WikiLink" title="Falcon 9 v1.0">F9 v1.0</a><sup about="#mw795" class="mw-ref reference" data-mw="{ "name": "ref", "attrs": { "name": "MuskMay2012" } }' id="cite_ref-MuskMay2012-13-1" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-MuskMay2012-13" id="mw7w"><span class="mw-reflink-text" id="mw8A"><span class="cite-bracket" id="mw8Q"></span>7<span class="cite-bracket" id="mw8g"></span></span></a></sup><br id="mw8w"/>B0004.1<sup about="#mw796" class="mw-ref reference" data-mw="{ "name": "ref", "attrs": { "name": "block_numbers" } }' id="cite_ref-block_numbers_14-1" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-block_numbers-14" id="mw9A"><span class="mw-reflink-text" id="mw9Q"><span class="cite-bracket" id="mw9g"></span>8<span class="cite-bracket" id="mw9w"></span></span></a></sup></td>
<td id="mw-A"><a href="https://en.wikipedia.org/wiki/Cape_Canaveral_Space_Force_Station" id="mw-Q" rel="mw:WikiLink" title="Cape Canaveral Space Force Station">CCAFS</a>,<br id="mw-g"/><a href="https://en.wikipedia.org/wiki/Cape_Canaveral_Space_Launch_Complex_40" id="mw-w" rel="mw:WikiLink" title="Cape Canaveral Space Launch Complex 40">SLC-40</a></td>
<td id="mw_A"><a href="https://en.wikipedia.org/wiki/SpaceX_Dragon" id="mw_Q" rel="mw:WikiLink" title="SpaceX Dragon">Dragon</a> <a class="mw-redirect" href="https://en.wikipedia.org/wiki/COTS_Demo_Flight_1" id="mw_g" rel="mw:WikiLink" title="COTS Demo Flight 1">demo flight C1</a><br id="mw_w"/>(Dragon C101)</td>
<td id="mwAQA"></td>
<td id="mwAQE"><a href="https://en.wikipedia.org/wiki/Low_Earth_orbit" id="mwAQI" rel="mw:WikiLink" title="Low Earth orbit">LEO</a> (<a href="https://en.wikipedia.org/wiki/International_Space_Station" id="mwAQM" rel="mw:WikiLink" title="International Space Station">ISS</a>)</td>
<td id="mwAQQ"><span about="#mw797" class="mw-empty-elt" data-mw="{ "parts": [ { "template": { "target": { "wt": "plainlist", "href": ". /Template:Plainlist", "params": { "1": { "wt": "\n* [[NASA]] ([[Commercial Orbital Transportation Services|COTS]])\n* [[National Reconnaissance Office|NRO]]\n" } }, "i": 0 } } ] }' id="mwAQU" typeof="mw:Transclusion"><style about="#mw798" data-mw="{ "name": "templatestyles", "attrs": { "src": "Plainlist/style.s.css" } }' data-mw-deduplicate="TemplateStyles:r1126788409" typeof="mw:Extension/templatestyles">.mw-parser-output .plainlist ol,.mw-parser-output .plainlist ul{line-height:inherit;list-style:none;margin:0;padding:0}.mw-parser-output .plainlist ol li,.mw-parser-output .plainlist ul li{margin-bottom:0}</style></span><div about="#mw797" class="plainlist" id="mwAQY">
<ul><li><a href="https://en.wikipedia.org/wiki/NASA" rel="mw:WikiLink" title="NASA">NASA</a> (<a href="https://en.wikipedia.org/wiki/Commercial_Orbital_Transportation_Services" rel="mw:WikiLink" title="Commercial Orbital Transportation Services">COTS</a>)</li>
<li><a href="https://en.wikipedia.org/wiki/National_Reconnaissance_Office" rel="mw:WikiLink" title="National Reconnaissance Office">NRO</a></li></ul>
</div></td>
<td about="#mw799" class="table-success" data-mw="{ "parts": [ " ", { "template": { "target": { "wt": "Success", "href": ". /Template:Success", "params": { }, "i": 0 } }, "&lt;ref name=\n\"ns20110930\"/&gt;"] }' id="mwAQc" style="background: #9EFF9E; color: black; vertical-align: middle; text-align: center; " typeof="mw:Transclusion">Success<sup about="#mw7100" class="mw-ref reference" data-mw="{ "name": "ref", "attrs": { "name": "ns20110930" } }' id="cite_ref-ns20110930_15-1" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-ns20110930-15" id="mwAQg"><span class="mw-reflink-text" id="mwAQk"><span class="cite-bracket" id="mwAQo"></span>9<span class="cite-bracket" id="mwAQ"

```

```

s">]</span></span></a></sup></td>
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<tr id="mwARU">
<td colspan="9" id="mwARY">Maiden flight of <a class="mw-redirect" href="https://en.wikipedia.org/wiki/Dragon_capsule" id="mwARc" rel="mw:WikiLink" title="Dragon capsule">Dragon capsule</a>, consisting of over 3 hours of testing thruster maneuvering and reentry.<sup about="#mwt108" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"spaceflightnow Clark unleashing Dragon\"},\"body\":{\"id\":\"mw-reference-text-cite_note-spaceflightnow_Clark_unleashing_Dragon-21\"}}\" id="cite_ref-spaceflightnow_Clark_unleashing_Dragon_21-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-spaceflightnow_Clark_unleashing_Dragon-21" id="mwARg"><span class="mw-reflink-text" id="mwARK"><span class="cite-bracket" id="mwARo">[</span>15<span class="cite-bracket" id="mwARs"></span></span></a></sup> Attempted to recover the first stage by parachuting it into the ocean, but it disintegrated upon reentry, before the parachutes were deployed.<sup about="#mwt109" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"parachute\"}}\" id="cite_ref-parachute_18-1" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-parachute-18" id="mwARw"><span class="mw-reflink-text" id="mwAR0"><span class="cite-bracket" id="mwAR4">[</span>12<span class="cite-bracket" id="mwAR8"></span></span></a></sup> <small id="mwASA">(<a class="mw-selflink-fragment" href="#COTS_demo_missions" id="mwASE" rel="mw:WikiLink">more details below</a></small> It also included two <a href="https://en.wikipedia.org/wiki/CubeSat" id="mwASI" rel="mw:WikiLink" title="CubeSat">CubeSats</a>,<sup about="#mwt112" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"NRO Taps Boeing for Next Batch of CubeSats\"},\"body\":{\"id\":\"mw-reference-text-cite_note-NRO_Taps_Boeing_for_Next_Batch_of_CubeSats-22\"}}\" id="cite_ref-NRO_Taps_Boeing_for_Next_Batch_of_CubeSats_22-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-NRO_Taps_Boeing_for_Next_Batch_of_CubeSats-22" id="mwASM"><span class="mw-reflink-text" id="mwASQ"><span class="cite-bracket" id="mwASU">[</span>16<span class="cite-bracket" id="mwASY"></span></span></a></sup> and a wheel of <a href="https://en.wikipedia.org/wiki/Brouère" id="mwAsc" rel="mw:WikiLink" title="Brouère">Brouère</a> cheese.</td></tr>
<tr id="mwASg">
<th id="mwASK" rowspan="2" scope="row" style="text-align:center;">3</th>
<td id="mwASo">22 May 2012,<br id="mwASs"/>07:44<sup about="#mwt113" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"BBC new era\"}}\" id="cite_ref-BBC_new_era_23-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-BBC_new_era-23" id="mwASw"><span class="mw-reflink-text" id="mwAS0"><span class="cite-bracket" id="mwAS4">[</span>17<span class="cite-bracket" id="mwAS8"></span></span></a></sup></td>
<td id="mwATA"><a href="https://en.wikipedia.org/wiki/Falcon_9_v1.0" id="mwATE" rel

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10/24

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enter; " typeof="mw:Transclusion">No<span typeof="mw:Entity"> </span>attempt</td></t
r>
<tr id="mwAVk">
<td colspan="9" id="mwAVo">Dragon spacecraft demonstrated a series of tests before i
t was allowed to approach the <a href="https://en.wikipedia.org/wiki/International_S
pace_Station" id="mwAVs" rel="mw:WikiLink" title="International Space Station">Inter
national Space Station</a>. Two days later, it became the first commercial spacecraf
t to board the ISS.<sup about="#mwt131" class="mw-ref reference" data-mw="{"name":"r
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_era-23"}}}" id="cite_ref-BBC_new_era_23-1" rel="dc:references" typeof="mw:Extension/r
ef"><a href="#cite_note-BBC_new_era-23" id="mwAVw"><span class="mw-reflink-text" id
="mwAV0"><span class="cite-bracket" id="mwAV4"></span>17<span class="cite-bracket"
id="mwAV8"></span></span></a></sup> <small id="mwAWA"><a class="mw-selflink-fragme
nt" href="#COTS_demo_missions" id="mwAWE" rel="mw:WikiLink">more details below</a>
</small></td></tr>
<tr id="mwAWI">
<th id="mwAWM" rowspan="3" scope="row" style="text-align:center;">4</th>
<td id="mwAWQ" rowspan="2">8 October 2012,<br id="mwAWU"/>00:35<sup about="#mwt132"
class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"SFN LLog"}}}" id="ci
te_ref-SFN_LLog_27-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_n
ote-SFN_LLog-27" id="mwAWY"><span class="mw-reflink-text" id="mwAWc"><span class="ci
te-bracket" id="mwAWg"></span>21<span class="cite-bracket" id="mwAWk"></span></spa
n></a></sup></td>
<td id="mwAWo" rowspan="2"><a href="https://en.wikipedia.org/wiki/Falcon_9_v1.0" id
="mwAWs" rel="mw:WikiLink" title="Falcon 9 v1.0">F9 v1.0</a><sup about="#mwt133" cla
ss="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"MuskMay2012"}}}" id="ci
te_ref-MuskMay2012_13-3" rel="dc:references" typeof="mw:Extension/ref"><a href="#cit
e_note-MuskMay2012-13" id="mwAWw"><span class="mw-reflink-text" id="mwAW0"><span cla
ss="cite-bracket" id="mwAW4"></span>7<span class="cite-bracket" id="mwAW8"></span>
</span></a></sup><br id="mwAXA"/>B0006.1<sup about="#mwt134" class="mw-ref referenc
e" data-mw="{"name":"ref","attrs":{"name":"block_numbers"}}}" id="cite_ref-block_numb
ers_14-3" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-block_nu
mbers-14" id="mwAXE"><span class="mw-reflink-text" id="mwAXI"><span class="cite-brac
ket" id="mwAXM"></span>8<span class="cite-bracket" id="mwAXQ"></span></span></a></
sup></td>
<td id="mwAXU" rowspan="2"><a href="https://en.wikipedia.org/wiki/Cape_Canaveral_Spa
ce_Force_Station" id="mwAXY" rel="mw:WikiLink" title="Cape Canaveral Space Force Sta
tion">CCAFS</a>,<br id="mwAXc"/><a href="https://en.wikipedia.org/wiki/Cape_Canavera
l_Space_Launch_Complex_40" id="mwAXg" rel="mw:WikiLink" title="Cape Canaveral Space
Launch Complex 40">SLC-40</a></td>
<td id="mwAXk"><a href="https://en.wikipedia.org/wiki/SpaceX_CRS-1" id="mwAXo" rel
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ef reference" data-mw="{"name":"ref","attrs":{"name":"sxManifest20120925"},"body":
{"id":"mw-reference-text-cite_note-sxManifest20120925-28"}}}" id="cite_ref-sxManifest
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anifest20120925-28" id="mwAXs"><span class="mw-reflink-text" id="mwAXw"><span class
="cite-bracket" id="mwAX0"></span>22<span class="cite-bracket" id="mwAX4"></span>
</span></a></sup><br id="mwAX8"/>(Dragon C103)</td>
<td id="mwAYA"><span about="#mwt138" data-mw="{"parts":[{"template":{"target":{"w
t":"cvt","href":"./Template:Cvt"},"params":{"1":{"wt":"4700"},"2":{"wt":"kg"}},"i":
0}}}]}" id="mwAYE" typeof="mw:Transclusion">4,700</span><span about="#mwt138" typeof
="mw:Entity"> </span><span about="#mwt138">kg (10,400</span><span about="#mwt138" ty
peof="mw:Entity"> </span><span about="#mwt138">lb)</span></td>
<td id="mwAYI"><a href="https://en.wikipedia.org/wiki/Low_Earth_orbit" id="mwAYM" re

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l="mw:WikiLink" title="Low Earth orbit">LEO</a> (<a href="https://en.wikipedia.org/wiki/International_Space_Station" id="mwAYQ" rel="mw:WikiLink" title="International Space Station">ISS</a>)</td>
<td id="mwAYU"><a href="https://en.wikipedia.org/wiki/NASA" id="mwAYY" rel="mw:WikiLink" title="NASA">NASA</a> (<a href="https://en.wikipedia.org/wiki/Commercial_Resupply_Services" id="mwAYc" rel="mw:WikiLink" title="Commercial Resupply Services">CRS</a>)</td>
<td about="#mwt139" class="table-success" data-mw="{"parts":["|","{"template":{"target":{"wt":"Success","href":"./Template:Success"},"params":{},"i":0}}]}' id="mwAYg" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: center; " typeof="mw:Transclusion">Success</td>
<td id="mwAYk" rowspan="2" style="background:#ecec; text-align:center;"><span about="#mwt140" class="nowrap" data-mw="{"parts":["{"template":{"target":{"wt":"nobr","href":"./Template:Nobr"},"params":{"1":{"wt":"No attempt"}},"i":0}}]}' id="mwAYo" typeof="mw:Transclusion">No attempt</span></td></tr>
<tr id="mwAYS">
<td id="mwAYw"><a href="https://en.wikipedia.org/wiki/Orbcomm_(satellite)" id="mwAY0" rel="mw:WikiLink" title="Orbcomm (satellite)">Orbcomm-OG2</a><sup about="#mwt143" class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"Orbcomm"},"body":{"id":"mw-reference-text-cite_note-Orbcomm-29"}}' id="cite_ref-Orbcomm_29-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-Orbcomm-29" id="mwAY4"><span class="mw-reflink-text" id="mwAY8"><span class="cite-bracket" id="mwAZA">[</span>23<span class="cite-bracket" id="mwAZE">]</span></span></a></sup></td>
<td id="mwAZI"><span about="#mwt144" data-mw="{"parts":["{"template":{"target":{"wt":"cvt","href":"./Template:Cvt"},"params":{"1":{"wt":"172"},"2":{"wt":"kg"}},"i":0}}]}' id="mwAZM" typeof="mw:Transclusion">172</span><span about="#mwt144" typeof="mw:Entity"></span><span about="#mwt144">kg (379</span><span about="#mwt144" typeof="mw:Entity"></span><span about="#mwt144">lb)</span><sup about="#mwt145" class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"gunter-og2"}}' id="cite_ref-gunter-og2_30-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-gunter-og2-30" id="mwAZQ"><span class="mw-reflink-text" id="mwAZU"><span class="cite-bracket" id="mwAZY">[</span>24<span class="cite-bracket" id="mwAZc">]</span></span></a></sup></td>
<td id="mwAZg"><a href="https://en.wikipedia.org/wiki/Low_Earth_orbit" id="mwAZk" rel="mw:WikiLink" title="Low Earth orbit">LEO</a></td>
<td id="mwAZo"><a href="https://en.wikipedia.org/wiki/Orbcomm" id="mwAZs" rel="mw:WikiLink" title="Orbcomm">Orbcomm</a></td>
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<tr id="mwAaE">
<td colspan="9" id="mwAaI">CRS-1 was successful, but the <a href="https://en.wikipedia.org/wiki/Secondary_payload" id="mwAaM" rel="mw:WikiLink" title="Secondary payload">secondary payload</a> was inserted into an abnormally low orbit and subsequently lost. This was due to one of the nine <a href="https://en.wikipedia.org/wiki/SpaceX_Merlin" id="mwAaQ" rel="mw:WikiLink" title="SpaceX Merlin">Merlin engines</a> shutting down during the launch, and NASA declining a second reignition, as per <a href="https://en.wikipedia.org/wiki/International_Space_Station" id="mwAaU" rel="mw:WikiLink" title="International Space Station">ISS</a> visiting vehicle safety rules, the primary payload owner is contractually allowed to decline a second reignition. NASA st

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ated that this was because SpaceX could not guarantee a high enough likelihood of the a second stage completing the second burn successfully which was required to avoid a ny risk of secondary payload's collision with the ISS.^{26}^{27}^{28}</td></tr>

<tr id="mwAbI">

<th id="mwAbM" rowspan="2" scope="row" style="text-align:center;">5</th>

<td id="mwAbQ">1 March 2013,<br id="mwAbU"/>15:10</td>

<td id="mwAbY">F9 v1.0^{7}<br id="mwAbw"/>B0007.1^{8}</td>

<td id="mwAcE">CCAFS,<br id="mwAcM"/>SLC-40</td>

<td id="mwAcU">SpaceX CRS-2^{22}<br id="mwAcs"/>(Dragon C104)</td>

<td id="mwAcw">4,877kg (10,752lb)</td>

<td id="mwAc4">LEO (ISS)</td>

<td id="mwAdE">NASA (CRS

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a>></td>
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<tr id="mwAdY">
<td colspan="9" id="mwAdc">Last launch of the original Falcon 9 v1.0 <a href="http://en.wikipedia.org/wiki/Launch_vehicle" id="mwAdg" rel="mw:WikiLink" title="Launch vehicle">launch vehicle</a>, first use of the unpressurized trunk section of Dragon.
<sup about="#mwt161" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"sxf9_20110321\"}}\" id="cite_ref-sxf9_20110321_35-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-sxf9_20110321-35" id="mwAdk"><span class="mw-reflink-text" id="mwAdo"><span class="cite-bracket" id="mwAds">[</span>29<span class="cite-bracket" id="mwAdw">]</span></span></a></sup></td></tr>
<tr id="mwAd0">
<th id="mwAd4" rowspan="2" scope="row" style="text-align:center;">6</th>
<td id="mwAd8">29 September 2013,<br id="mwAeA"/>16:00<sup about="#mwt162" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"pa20130930\"}}\" id="cite_ref-pa20130930_36-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-pa20130930-36" id="mwAeE"><span class="mw-reflink-text" id="mwAeI"><span class="cite-bracket" id="mwAeM">[</span>30<span class="cite-bracket" id="mwAeQ">]</span></span></a></sup></td>
<td id="mwAeU"><a href="https://en.wikipedia.org/wiki/Falcon_9_v1.1" id="mwAeY" rel="mw:WikiLink" title="Falcon 9 v1.1">F9 v1.1</a><sup about="#mwt163" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"MuskMay2012\"}}\" id="cite_ref-MuskMay2012_13-5" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-MuskMay2012-13" id="mwAec"><span class="mw-reflink-text" id="mwAeg"><span class="cite-bracket" id="mwAek">[</span>7<span class="cite-bracket" id="mwAeo">]</span></span></a></sup><br id="mwAes"/>B1003<sup about="#mwt164" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"block_numbers\"}}\" id="cite_ref-block_numbers_14-5" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-block_numbers-14" id="mwAew"><span class="mw-reflink-text" id="mwAe0"><span class="cite-bracket" id="mwAe4">[</span>8<span class="cite-bracket" id="mwAe8">]</span></span></a></sup></td>
<td id="mwAfA"><a class="mw-redirect" href="https://en.wikipedia.org/wiki/Vandenberg_Air_Force_Base" id="mwAfE" rel="mw:WikiLink" title="Vandenberg Air Force Base">VAFB</a>,<br id="mwAfI"/><a href="https://en.wikipedia.org/wiki/Vandenberg_Space_Launch_Complex_4" id="mwAfM" rel="mw:WikiLink" title="Vandenberg Space Launch Complex 4">SLC-4E</a></td>
<td id="mwAfQ"><a href="https://en.wikipedia.org/wiki/CASSIOPE" id="mwAfU" rel="mw:WikiLink" title="CASSIOPE">CASSIOPE</a><sup about="#mwt165" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"sxManifest20120925\"}}\" id="cite_ref-sxManifest20120925_28-2" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-sxManifest20120925-28" id="mwAfY"><span class="mw-reflink-text" id="mwAfc"><span class="cite-bracket" id="mwAfg">[</span>22<span class="cite-bracket" id="mwAfk">]</span></span></a></sup><sup about="#mwt168" class="mw-ref reference" data-mw="{\"name\":\"ref\",\"attrs\":{\"name\":\"CASSIOPE_MDA\"},\"body\":{\"id\":\"mw-reference-text-cite_note-CASSIOPE_MDA-37\"}}\" id="cite_ref-CASSIOPE_MDA_37-0" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-CASSIOPE_MDA-37" id="mwAfo"><span class="mw-reflink-text" id="mwAfs"><span class="cite-bracket" id="mwAfw">[</span>31<span class="cite-bracket" id="mwAf0">]</span></span></a></sup></td>
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0}}}]' id="mwAf8" typeof="mw:Transclusion">500</span><span about="#mwt169" typeof="m
w:Entity"> </span><span about="#mwt169">kg (1,100</span><span about="#mwt169" typeof
="mw:Entity"> </span><span about="#mwt169">lb)</span></td>
<td id="mwAgA"><a href="https://en.wikipedia.org/wiki/Polar_orbit" id="mwAgE" rel="m
w:WikiLink" title="Polar orbit">Polar orbit</a> <a href="https://en.wikipedia.org/wi
ki/Low_Earth_orbit" id="mwAgI" rel="mw:WikiLink" title="Low Earth orbit">LEO</a></td
>
<td id="mwAgM"><a href="https://en.wikipedia.org/wiki/Maxar_Technologies" id="mwAgQ"
rel="mw:WikiLink" title="Maxar Technologies">MDA</a></td>
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a20130930&gt;&gt;"]}" id="mwAgU" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center; " typeof="mw:Transclusion">Success<sup about="#mwt17
1" class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"pa20130930"}}" i
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</span></span></a></sup></td>
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lt;br/&gt;&lt;small&gt;(ocean)&lt;/small&gt;{{efn|name=ocean landing}}}" id="mwAg
o" style="background: #FFE3E3; color: black; vertical-align: middle; text-align: cen
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mwt173" class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"ocean landi
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id="mwAgs"><span class="mw-reflink-text" id="mwAgw"><span class="cite-bracket" id="m
wAg0"></span>d<span class="cite-bracket" id="mwAg4"></span></span></a></sup></td>
</tr>
<tr id="mwAg8">
<td colspan="9" id="mwAhA">First commercial mission with a private customer, first l
aunch from Vandenberg, and demonstration flight of Falcon 9 v1.1 with an improved 13
-tonne to LEO capacity.<sup about="#mwt178" class="mw-ref reference" data-mw="{"nam
e":"ref","attrs":{"name":"sxf9_20110321"},"body":{"id":"mw-reference-text-cite_note-
sxf9_20110321-35"}}" id="cite_ref-sxf9_20110321_35-1" rel="dc:references" typeof="m
w:Extension/ref"><a href="#cite_note-sxf9_20110321-35" id="mwAhE"><span class="mw-re
flink-text" id="mwAhI"><span class="cite-bracket" id="mwAhM"></span>29<span class
="cite-bracket" id="mwAhQ"></span></span></a></sup> After separation from the secon
d stage carrying Canadian commercial and scientific satellites, the first stage boos
ter performed a controlled reentry,<sup about="#mwt181" class="mw-ref reference" dat
a-mw="{"name":"ref","attrs":{},"body":{"id":"mw-reference-text-cite_note-39"}}" id
="cite_ref-39" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-39"
id="mwAhU"><span class="mw-reflink-text" id="mwAhY"><span class="cite-bracket" id="m
wAhc"></span>32<span class="cite-bracket" id="mwAhg"></span></span></a></sup> and
an <a href="https://en.wikipedia.org/wiki/Falcon_9_first-stage_landing_tests" id="mw
Ahk" rel="mw:WikiLink" title="Falcon 9 first-stage landing tests">ocean touchdown te
st</a> for the first time. This provided good test data, even though the booster sta
rted rolling as it neared the ocean, leading to the shutdown of the central engine a
s the roll depleted it of fuel, resulting in a hard impact with the ocean.<sup about
="#mwt182" class="mw-ref reference" data-mw="{"name":"ref","attrs":{"name":"pa201309
30"}}" id="cite_ref-pa20130930_36-2" rel="dc:references" typeof="mw:Extension/ref"><
a href="#cite_note-pa20130930-36" id="mwAho"><span class="mw-reflink-text" id="mwAh
s"><span class="cite-bracket" id="mwAhw"></span>30<span class="cite-bracket" id="mw

```

Ah0"></sup>

This was the first known attempt of a rocket engine b eing lit to perform a supersonic retro propulsion, and allowed SpaceX to enter a pub lic-private partnership with NASA and its Mars entry, descent, and landing tec hnologies research projects.^{33}

<small id="mwAiM">more details below</small></td></tr>

<tr id="mwAiU">

<th id="mwAiY" rowspan="2" scope="row" style="text-align:center;">7</th>

<td id="mwAic">3 December 2013,<br id="mwAig"/>22:41^{34}</td>

<td id="mwAi0">F9 v1.1<br id="mwAi8"/>B1004</td>

<td id="mwAjA">CCAFS ,<br id="mwAjI"/>SLC-40</td>

<td id="mwAjQ">SES-8^{22}^{35}^{36</spa n>}</td>

<td id="mwAKI">3,170kg (6,990lbf)</td>

<td id="mwAKq">GTO</td>

<td id="mwAKy">SES</td>

<td about="#mwt198" class="table-success" data-mw='{"parts":["| ", {"template":{"targ et":{"wt":"Success","href":"/Template:Success"},"params":{"},"i":0}}, "<ref name=S NMissionStatus7>{{cite web|title=SpaceflightNow Mission Status Center|url=http://

```

spaceflightnow.com/falcon9/007/status.html|publisher=Spaceflight Now|date=December
3, 2013 |archive-url=https://web.archive.org/web/20140328062838/https://spaceflightn
ow.com/falcon9/007/status.html|archive-date=March 28, 2014}}</ref>"]}' id="mwAk
kg" style="background: #9EFF9E; color:black; vertical-align: middle; text-align: cen
ter; " typeof="mw:Transclusion">Success<sup about="#mwt201" class="mw-ref reference"
data-mw='{"name":"ref","attrs":{"name":"SNMissionStatus7"},"body":{"id":"mw-referenc
e-text-cite_note-SNMissionStatus7-44"}}' id="cite_ref-SNMissionStatus7_44-0" rel="d
c:references" typeof="mw:Extension/ref"><a href="#cite_note-SNMissionStatus7-44" id
="mwAkK"><span class="mw-reflink-text" id="mwAkO"><span class="cite-bracket" id="mwa
ks"></span>37<span class="cite-bracket" id="mwAkW"></span></span></a></sup></td>
<td about="#mwt202" class="table-noAttempt" data-mw='{"parts":["| ",{"template":{"ta
rget":{"wt":"No attempt","href":"./Template:No_attempt"},"params":{},"i":0}},"&lt;b
r/&gt;&lt;ref name=\\"sf10120131203\\"&gt;{{cite web|title=SpaceX Falcon 9 v1.1 - SES-
8 Launch Updates|url=http://www.spaceflight101.net/spacex-falcon-9-v11-ses-8-launch-
updates.html|publisher=Spaceflight 101|date=December 3, 2013|access-date=July 13, 20
16|archive-url=https://web.archive.org/web/20160304034110/http://www.spaceflight101.
net/spacex-falcon-9-v11-ses-8-launch-updates.html|archive-date=March 4, 2016 |url-st
atus=dead}}</ref>"]}' id="mwAk0" style="background: #EEE; color:black; vertica
l-align: middle; text-align: center; " typeof="mw:Transclusion">No<span typeof="mw:E
ntity"> </span>attempt<br/><sup about="#mwt205" class="mw-ref reference" data-mw
='{"name":"ref","attrs":{"name":"sf10120131203"},"body":{"id":"mw-reference-text-cit
e_note-sf10120131203-45"}}' id="cite_ref-sf10120131203_45-0" rel="dc:references" typ
eof="mw:Extension/ref"><a href="#cite_note-sf10120131203-45" id="mwAk4"><span class
="mw-reflink-text" id="mwAk8"><span class="cite-bracket" id="mwAlA"></span>38<span
class="cite-bracket" id="mwAlE"></span></span></a></sup></td></tr>
<tr id="mwAlI">
<td colspan="9" id="mwAlM">First <a href="https://en.wikipedia.org/wiki/Geostationar
y_transfer_orbit" id="mwAlQ" rel="mw:WikiLink" title="Geostationary transfer orbit">
Geostationary transfer orbit</a> (GTO) launch for Falcon 9,<sup about="#mwt207" clas
s="mw-ref reference" data-mw='{"name":"ref","attrs":{"name":"spx-pr"}}' id="cite_ref
-spx-pr_42-1" rel="dc:references" typeof="mw:Extension/ref"><a href="#cite_note-spx-
pr-42" id="mwAlU"><span class="mw-reflink-text" id="mwAlY"><span class="cite-bracke
t" id="mwAlc"></span>35<span class="cite-bracket" id="mwAlg"></span></span></a></s
up> and first successful reignition of the second stage.<sup about="#mwt210" class
="mw-ref reference" data-mw='{"name":"ref","attrs":{},"body":{"id":"mw-reference-tex
t-cite_note-46"}}' id="cite_ref-46" rel="dc:references" typeof="mw:Extension/ref"><a
href="#cite_note-46" id="mwAlk"><span class="mw-reflink-text" id="mwAlO"><span class
="cite-bracket" id="mwAls"></span>39<span class="cite-bracket" id="mwAlw"></span>
</span></a></sup> SES-8 was inserted into a <a href="https://en.wikipedia.org/wiki/G
eostationary_transfer_orbit" id="mwAl0" rel="mw:WikiLink" title="Geostationary trans
fer orbit">Super-Synchronous Transfer Orbit</a> of <span about="#mwt206" data-mw
='{"parts":[{"template":{"target":{"wt":"cvt"},"href":"./Template:Cvt"},"params":
{"1":{"wt":"79341"},"2":{"wt":"km"}},{"i":0}}]' id="mwAl4" typeof="mw:Transclusion">
79,341</span><span about="#mwt206" typeof="mw:Entity"> </span><span about="#mwt206">
km (49,300</span><span about="#mwt206" typeof="mw:Entity"> </span><span about="#mwt2
06">mi)</span> in apogee with an <a href="https://en.wikipedia.org/wiki/Orbital_incl
ination" id="mwAl8" rel="mw:WikiLink" title="Orbital inclination">inclination</a> of
20.55° to the <a href="https://en.wikipedia.org/wiki/Equator" id="mwAmA" rel="mw:Wik
iLink" title="Equator">equator</a>.</td></tr>
</tbody></table>

```

You should be able to see the column names embedded in the table header elements `<th>` as follows:

```

<tr>
<th scope="col">Flight No.
</th>
<th scope="col">Date and<br/>time (<a
href="/wiki/Coordinated_Universal_Time" title="Coordinated
Universal Time">UTC</a>)
</th>
<th scope="col"><a href="/wiki/List_of_Falcon_9_first-
stage_boosters" title="List of Falcon 9 first-stage
boosters">Version,<br/>Booster</a> <sup class="reference"
id="cite_ref-booster_11-0"><a href="#cite_note-booster-11">[b]</a>
</sup>
</th>
<th scope="col">Launch site
</th>
<th scope="col">Payload<sup class="reference" id="cite_ref-
Dragon_12-0"><a href="#cite_note-Dragon-12">[c]</a></sup>
</th>
<th scope="col">Payload mass
</th>
<th scope="col">Orbit
</th>
<th scope="col">Customer
</th>
<th scope="col">Launch<br/>outcome
</th>
<th scope="col"><a href="/wiki/Falcon_9_first-stage_landing_tests"
title="Falcon 9 first-stage landing tests">Booster<br/>landing</a>
</th></tr>

```

Next, we just need to iterate through the `<th>` elements and apply the provided `extract_column_from_header()` to extract column name one by one

```

In [10]: column_names = []

# Apply find_all() function with `th` element on first_launch_table
# Iterate each th element and apply the provided extract_column_from_header() to get
# Append the Non-empty column name (`if name is not None and len(name) > 0`) into a
column_names = []

for th in first_launch_table.find_all("th"):
    name = extract_column_from_header(th)
    if name is not None and len(name) > 0:
        column_names.append(name)

print(column_names)

```

```

['Flight No.', 'Date and time ( )', 'Launch site', 'Payload', 'Payload mass', 'Orbit', 'Customer', 'Launch outcome']

```

Check the extracted column names

```
In [11]: print(column_names)
```

```
['Flight No.', 'Date and time ( )', 'Launch site', 'Payload', 'Payload mass', 'Orbit', 'Customer', 'Launch outcome']
```

TASK 3: Create a data frame by parsing the launch HTML tables

We will create an empty dictionary with keys from the extracted column names in the previous task. Later, this dictionary will be converted into a Pandas dataframe

```
In [12]: launch_dict= dict.fromkeys(column_names)

# Remove an irrelevant column
del launch_dict['Date and time ( )']

# Let's initial the launch_dict with each value to be an empty list
launch_dict['Flight No.'] = []
launch_dict['Launch site'] = []
launch_dict['Payload'] = []
launch_dict['Payload mass'] = []
launch_dict['Orbit'] = []
launch_dict['Customer'] = []
launch_dict['Launch outcome'] = []
# Added some new columns
launch_dict['Version Booster']=[]
launch_dict['Booster landing']=[]
launch_dict['Date']=[]
launch_dict['Time']=[]
```

Next, we just need to fill up the `launch_dict` with launch records extracted from table rows.

Usually, HTML tables in Wiki pages are likely to contain unexpected annotations and other types of noises, such as reference links `B0004.1[8]` , missing values `N/A [e]` , inconsistent formatting, etc.

To simplify the parsing process, we have provided an incomplete code snippet below to help you to fill up the `launch_dict` . Please complete the following code snippet with TODOs or you can choose to write your own logic to parse all launch tables:

```
In [13]: extracted_row = 0
#Extract each table
for table_number,table in enumerate(soup.find_all('table',"wikitable plainrowheader")):
    # get table row
    for rows in table.find_all("tr"):
        #check to see if first table heading is as number corresponding to launch a
        if rows.th:
            if rows.th.string:
                flight_number=rows.th.string.strip()
```

```

        flag=flight_number.isdigit()
    else:
        flag=False
    #get table element
    row=rows.find_all('td')
    #if it is number save cells in a dictionary
    if flag:
        extracted_row += 1
        # Flight Number value
        # TODO: Append the flight_number into Launch_dict with key `Flight No.`
        #print(flight_number)
        datatimelist=date_time(row[0])

        # Date value
        # TODO: Append the date into Launch_dict with key `Date`
        date = datatimelist[0].strip(',')
        #print(date)

        # Time value
        # TODO: Append the time into Launch_dict with key `Time`
        time = datatimelist[1]
        #print(time)

        # Booster version
        # TODO: Append the bv into Launch_dict with key `Version Booster`
        bv=booster_version(row[1])
        if not(bv):
            bv=row[1].a.string
        print(bv)

        # Launch Site
        # TODO: Append the bv into Launch_dict with key `Launch Site`
        launch_site = row[2].a.string
        #print(launch_site)

        # Payload
        # TODO: Append the payload into Launch_dict with key `Payload`
        payload = row[3].a.string
        #print(payload)

        # Payload Mass
        # TODO: Append the payload_mass into Launch_dict with key `Payload mass`
        payload_mass = get_mass(row[4])
        #print(payload)

        # Orbit
        # TODO: Append the orbit into Launch_dict with key `Orbit`
        orbit = row[5].a.string
        #print(orbit)

        # Customer
        # TODO: Append the customer into Launch_dict with key `Customer`
        customer = row[6].a.string
        #print(customer)

        # Launch outcome

```

```
# TODO: Append the launch_outcome into launch_dict with key `Launch out  
launch_outcome = list(row[7].strings)[0]  
#print(launch_outcome)  
  
# Booster Landing  
# TODO: Append the launch_outcome into launch_dict with key `Booster La  
booster_landing = landing_status(row[8])  
#print(booster_landing)
```

22/24

F9 B4B1045.2
F9 B5
F9 B5349B1048[
F9 B5B1046.2
F9 B5[
F9 B5B1048.2
F9 B5B1047.2
F9 B5B1046.3268
F9 B5[
F9 B5[
F9 B5B1049.2
F9 B5B1048.3
F9 B5[]
F9 B5[
F9 B5B1049.3
F9 B5B1051.2
F9 B5B1056.2
F9 B5B1047.3
F9 B5
F9 B5[
F9 B5B1056.3
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5
F9 B5[
F9 B5
F9 B5
F9 B5
F9 B5B1058.2
F9 B5
F9 B5B1049.6
F9 B5
F9 B5B1060.2
F9 B5B1058.3
F9 B5B1051.6
F9 B5
F9 B5
F9 B5[
F9 B5
F9 B5 △[
F9 B5 △[
F9 B5 △
F9 B5 △
F9 B5
F9 B5B1051.8
F9 B5B1058.5

```

-----
AttributeError                                Traceback (most recent call last)
/tmp/ipykernel_366/354354048.py in <module>
     60         # Customer
     61         # TODO: Append the customer into launch_dict with key `Customer`
--> 62         customer = row[6].a.string
     63         #print(customer)
     64

AttributeError: 'NoneType' object has no attribute 'string'

```

After you have fill in the parsed launch record values into `launch_dict`, you can create a dataframe from it.

```
In [ ]: df = pd.DataFrame({ key:pd.Series(value) for key, value in launch_dict.items() })
```

We can now export it to a **CSV** for the next section, but to make the answers consistent and in case you have difficulties finishing this lab.

Following labs will be using a provided dataset to make each lab independent.

```
df.to_csv('spacex_web_scraped.csv', index=False)
```

Authors

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[Nayef Abou Tayoun](#)

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